

Shading-Aware AI System for Solar Investment, Health Monitoring, and Grid-Resilience in Sri Lanka

J26-DS-323

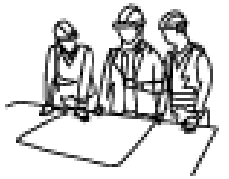
Shading-Aware AI System for Solar Investment, Health Monitoring, and Grid-Resilience in Sri Lanka

Who need this solution & Why?



Solar system Owners

- Panels underperform the quoted output — no way to know why
- Panel Degradation goes unnoticed until the bill spikes
- Free daytime solar power goes unused
- Battery is empty exactly when a blackout hits



Solar installers

- Siting decisions made by rule-of-thumb, no data to back the pitch
- Inspections are reactive — only after a complaint
- No ongoing value to offer after installation ends
- Only answer to resilience is "buy a bigger battery"

Proposed Solution

Solution Overview

Our solution is an integrated, AI-driven residential solar intelligence system for Sri Lankan households that supports the solar lifecycle from shading-aware installation optimization and panel health assessment, to solar-aware appliance scheduling and outage-resilient energy management.

Individual students' components/contributions

- Senadeera D.M.K.K - Vision-Based Shading Analysis and Solar Geometry Optimization: analyzes rooftop shading and panel geometry to recommend optimal tilt and orientation.
- Erandunu P.H.L - Vision-Based Solar Panel Health Assessment: diagnoses panel condition from a single photo, to identify degradation before it causes real power loss
- Perera P.K.M.P - Dynamic Solar-Aware Appliance Scheduling: forecasts day-ahead solar generation, optimizes flexible appliance schedules, and triggers rescheduling when actual solar generation significantly deviates from the forecast.
- Wijewardena P.K.L - Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning: decides hourly whether to self-consume, export, or reserve power, anticipating scheduled blackouts to keep critical loads running.

Vision Based Solar Panel Health Assessment



- Converts a single photo to estimate panel health score
- Outputs: degradation category, detected issue, estimated efficiency loss, recommendation
- Contribution: the panel-condition intelligence layer of the overall system

IT23429628 | Erandunu P.H.L | J26-DS-323

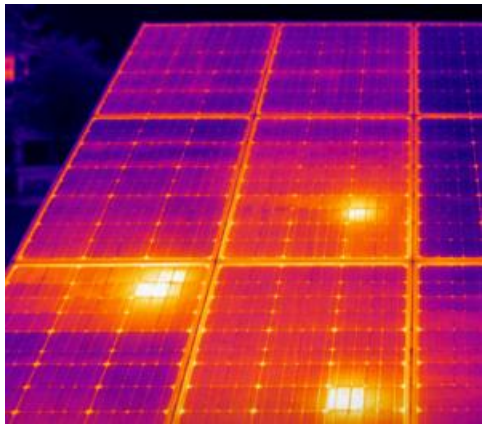
Novelty

Problem

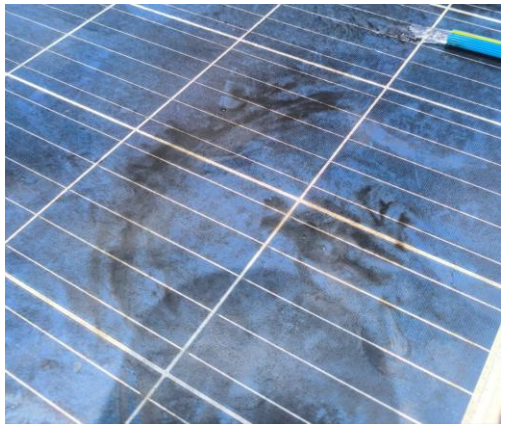
- Currently Panel monitoring today relies on manual visual inspection (inconsistent) or thermal/ electroluminescence imaging (costly, impractical for households)

- Existing Low-cost solutions are often trained on data from different climates, so they may not reflect in tropical countries like Sri Lanka.

- Most importantly, existing systems don't clearly separate surface issues from actual panel damage.



Thermal Imaging





Algae

Novelty

Research Gap

- Existing panel classifiers are trained on public datasets (dust, snow, bird-droppings, cracks)
- None target tropical degradation modes: algae growth and coastal salt-haze corrosion



Salt haze corrosion



Novelty

- **First, instead of only giving a healthy or damaged label, we provide a continuous health score linked to expected efficiency loss**
- **Surface problems and actual structural damage are reported separately, so a dirty panel is not treated as a damaged panel**



Novelty

Main Objective

Develop a computer-vision model that estimates panel degradation from photographs, calibrated for tropical conditions, and validated against real measured efficiency loss.

Sub-objectives

1. Collect and label field-image dataset across installation ages
2. Design and train a CNN model using two-stage transfer learning
3. Apply Grad-CAM explainability to show what areas influenced the model's decision
4. Implement tiered recommendation logic (Healthy / Early / Moderate / Severe) with structural vs. surface separation



Research Methodology



1.Data Preprocessing & Augmentation

Domain: Computer Vision / Image Processing
Methods: CLAHE lighting normalization, cropping, augmentation
Tools: Python, OpenCV,
Inputs: Field photos + public dataset (pretraining)

2. Two-Stage Transfer Learning

Domain: Deep Learning
Method: CNN (ResNet, EfficientNet) — classification pretrain, regression fine-tune
Tools: PyTorch,
Outputs: Labelled severity scores (0–100)

3. Explainability & Recommendation

Domain: Explainable AI / Decision Logic
Method: Grad-CAM + tiered threshold logic
Tools: pytorch-grad-cam
Inputs: Trained model output → health score
Output: recommendation logics

Research Methodology

Validation & Evaluation

- **Held-out test accuracy-** MAE / RMSE against labelled scores never seen during training
- **Real-world correlation-** Pearson correlation (or R^2) compared between predicted health score with actual efficiency loss from 10–15 panels the core evidence the score reflects real performance
- **Grad-CAM sanity check-** using Grad-CAM to verify the model focuses on the actual problem area

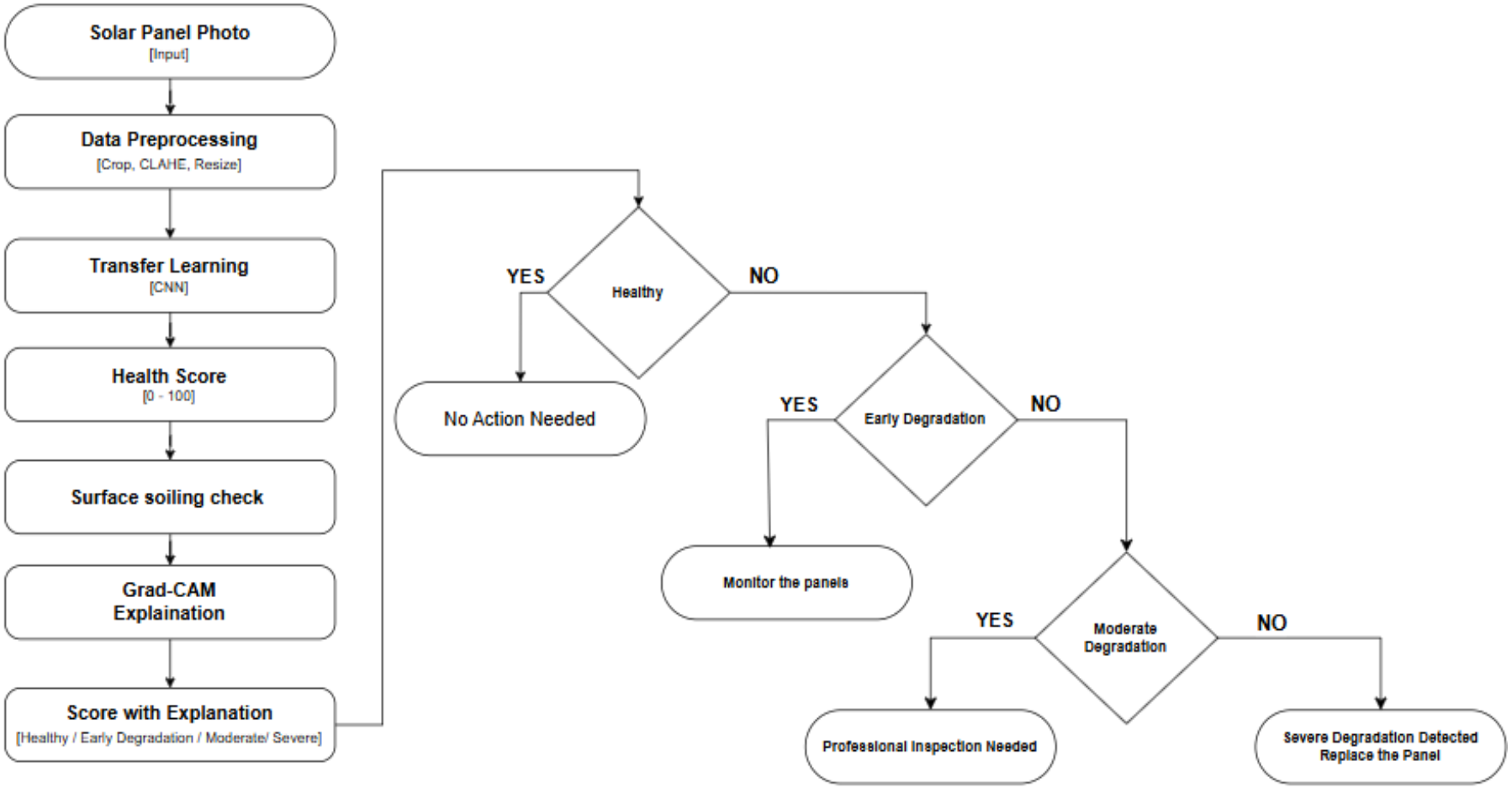
Data Availability, Security & Ethics

- **Sources:** first-hand field photography (target 80–150 panels, age-stratified) + public datasets for pretraining only
- **Access:** formal written request for site access, submitted through a solar installer
- **Consent:** obtained and logged before every site visit
- **Privacy:** no personal information collected — only the photo of panels and generation data

Vision Based Solar Panel Health Assessment

Implementation Details

System Overview



Implementation Details

Dependencies

- Requires only a standard smartphone photo — no sensors or specialized hardware





Dynamic Solar-Aware Appliance Scheduling

IT23191488 | Perera P.K.M.P

Novelty

Problem

- Solar generation is often high around midday, while household demand may be low.
- Excess solar is exported to the grid; high export during low grid demand can contribute to voltage rise and inverter curtailment.
- Flexible high-demand appliances could be shifted to high-solar periods to increase direct solar utilization.
- However, day-ahead scheduling depends on PV forecasts, and actual generation can change with weather.
- Therefore, the initial appliance schedule may become unsuitable during the day.



Novelty

Research Gap

Existing approaches use predicted solar generation for appliance scheduling, while some reschedule periodically. However, they may not determine when rescheduling is actually necessary when actual PV differs from the forecast.



Novelty: A dynamic appliance rescheduling mechanism that triggers re-optimization only when a significant deviation is detected between predicted and actual PV generation.

Novelty

Main Objective

Develop an adaptive solar aware appliance scheduling framework using PV forecasting and forecast deviation driven dynamic rescheduling.

Sub-objectives

1. Prepare PV, weather & solar-position data
2. Develop day-ahead PV forecasting
3. Model flexible appliance constraints
4. Develop MILP day-ahead scheduling
5. Develop deviation-triggered dynamic rescheduling
6. Compare against fixed, day-ahead & periodic methods



Research Methodology



1. PV Forecasting

Domain: Machine Learning / Data Science
Methods: XGBoost / Random Forest
Tools: Python, Pandas, scikit-learn, pvlib
Inputs: PV + weather + solar position

2. Appliance Scheduling

Domain: Mathematical Optimization
Method: Mixed-Integer Linear Programming (MILP)
Tools: Pyomo or PuLP Frameworks
Inputs: PV forecast + appliance constraints

3. Dynamic Rescheduling

Domain: Adaptive Energy Management
Method: Event / threshold trigger
Compare: Forecast PV ↔ Actual PV
Action: MILP re-optimization

Research Methodology





Validation & Evaluation

- PV Forecasting: MAE, RMSE,
- Scheduling: Solar-load overlap, appliance constraint satisfaction
- Dynamic Rescheduling: Solar-load overlap, trigger count, execution time / computational overhead
- Baseline Comparison: Fixed → Day-Ahead → Periodic → Proposed Event-Triggered

Research Methodology



Data Availability

- Historical residential PV generation (open-meteo.com)
- Historical + forecast weather (pvlib)
- Solar-position features
- User-defined appliance constraints
- Actual / replayed PV for rescheduling evaluation

Security

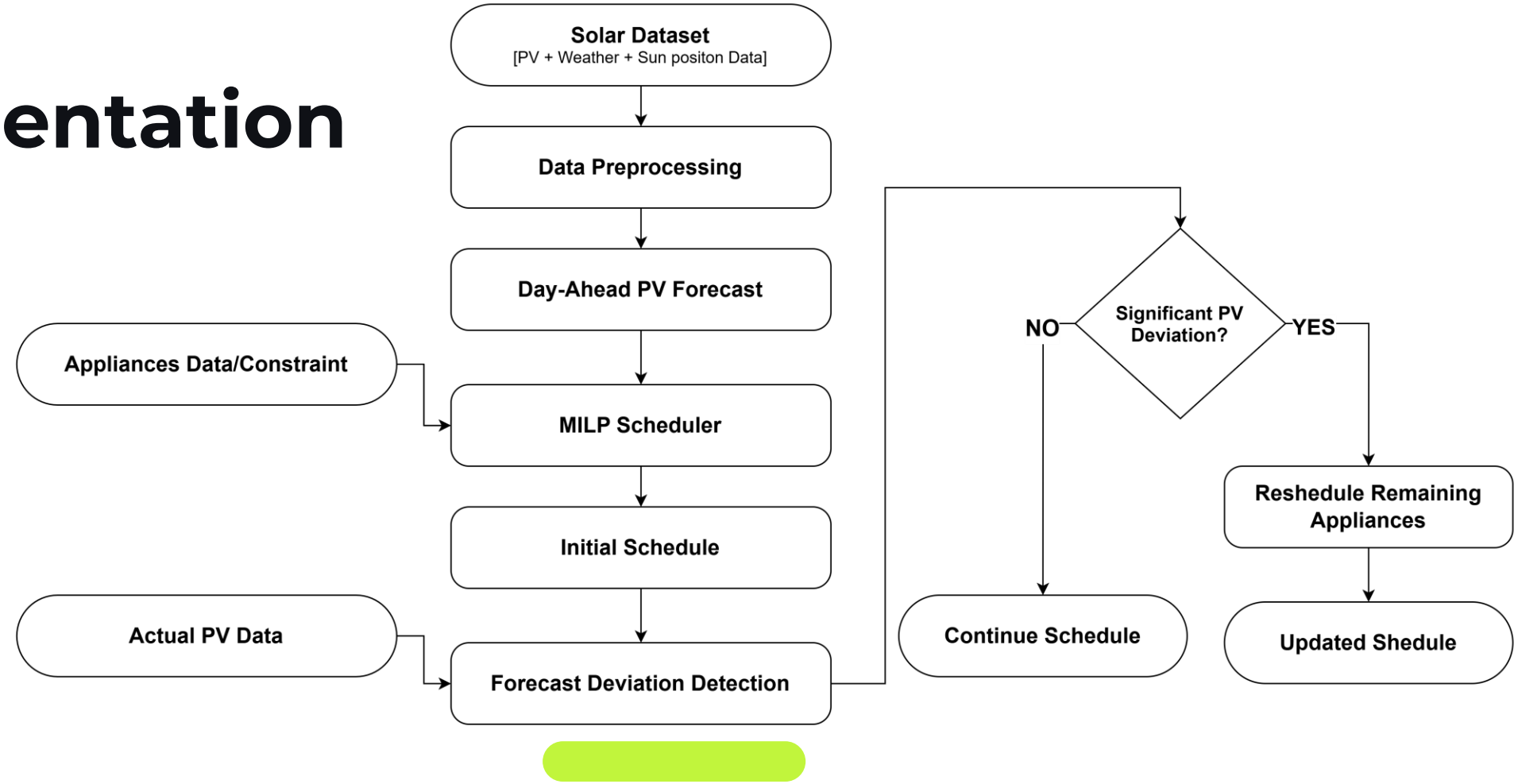
Anonymized system IDs
No Personal identified information required
Secure handling of energy data

Ethics

Anonymized user data; no personal identifiers
Consent for direct user data collection

Implementation Details

System Overview



Implementation Details

Key dependencies

- Forecasting → PV + weather + solar position
- Scheduler → forecast + appliance constraints
- Detector → forecast + actual PV
- Rescheduler → trigger + remaining constraints



Vision-Based Shading Analysis & Solar Geometry Optimization



IT23343320 | Senadeera D.M.K.K

Vision-Based Shading Analysis & Solar Geometry Optimization

Novelty

Problem



- Localized micro-shading (e.g., trees, chimneys) causes massive, disproportionate energy loss in residential PV strings.
- Current accurate shading assessments require cost-prohibitive physical hardware (Solmetric SunEye) or complex 3D LiDAR models.
- Local installations heavily rely on generic "rule-of-thumb" setups (e.g., fixed 7° tilt), completely ignoring site-specific structural shadows.
- Existing passive AI models only segment the sky but fail to output actionable installation geometries to actively avoid the shade.



Vision-Based Shading Analysis & Solar Geometry Optimization



Novelty

Research Gap

Existing shading analysis relies on inaccessible hardware or passive sky-segmentation models that fail to compute actionable, shade-evasive installation angles.

Novelty: Hardware-free active geometric optimization- integrates CNN fisheye segmentation, pvlib sun-paths, and PSO to dynamically compute actionable optimal tilt (5°-30°) and azimuth angles.



IT23343320 | Senadeera D.M.K.K | J26-DS-323

Novelty

Main Objective

Develop a hardware-free, AI-driven spatio-temporal shading analysis framework to dynamically optimize residential solar PV tilt (5°–30°) and azimuth angles using upward-facing fisheye imagery.

Sub Objective

1. Prepare and augment strictly North-aligned fisheye image datasets.
2. Develop a CNN model (DeepLabV3/U-Net++) for sky and obstacle segmentation.
3. Model annual solar trajectories using pvlib pixel-to-angle projection.
4. Develop a Particle Swarm Optimization (PSO) engine for angle calculation.
5. Evaluate the optimized energy yield against standard fixed-tilt installations.

Research Methodology

Sky-Obstacle Segmentation

Domain: Computer Vision / Deep Learning
Methods: Semantic Segmentation, Transfer Learning
Tools: Python, PyTorch/TensorFlow, DeepLabV3 / U-Net++
Inputs: 180° upward-facing fisheye imagery (Insta360)

Spatio-Temporal Mapping

Domain: Computer Vision / Deep Learning
Methods: Semantic Segmentation, Transfer Learning
Tools: Python, PyTorch/TensorFlow, DeepLabV3 / U-Net++
Inputs: 180° upward-facing fisheye imagery (Insta360)

Constrained Angle Optimization

Domain: Metaheuristic Optimization / Swarm Intelligence
Method: Particle Swarm Optimization (PSO)
Constraints: Hardware mounting limits (Tilt constrained to 5°–30°)
Output: Dynamically computed, shade-evasive optimal Tilt & Azimuth

Research Methodology





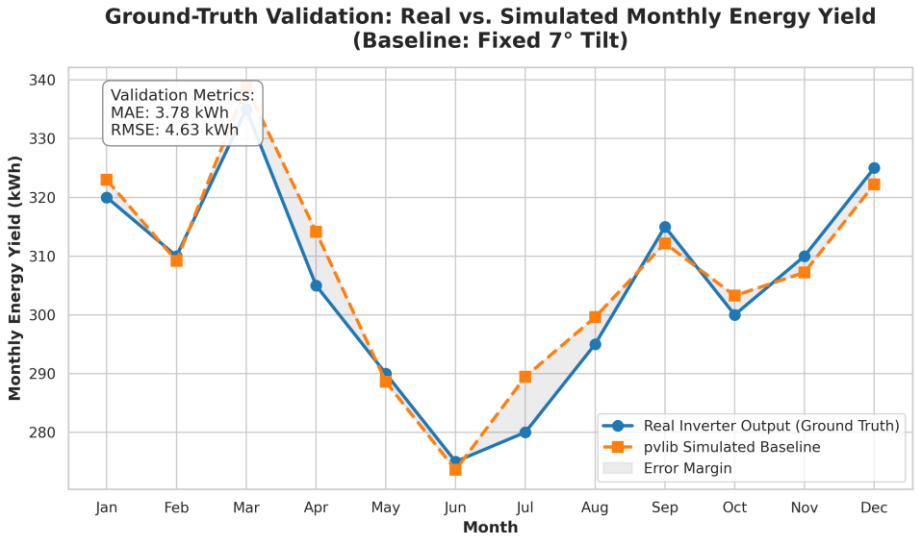


Data Availability, Security, and Ethical Clearance:

- **Data Sourcing:** Utilizing a localized, custom-captured dataset of 180° upward-facing fisheye images via consumer-grade Insta360 cameras.
- **Augmentation Strategy (Critical):** Dataset expanded employing strictly pixel-level augmentations (Brightness, Contrast, Noise) to rigorously preserve North-Alignment for valid sun-path math.
- **Security & Ethics:** Non-intrusive skyward captures ensure absolutely no Personally Identifiable Information (PII) or private property interiors are recorded.

Research Methodology

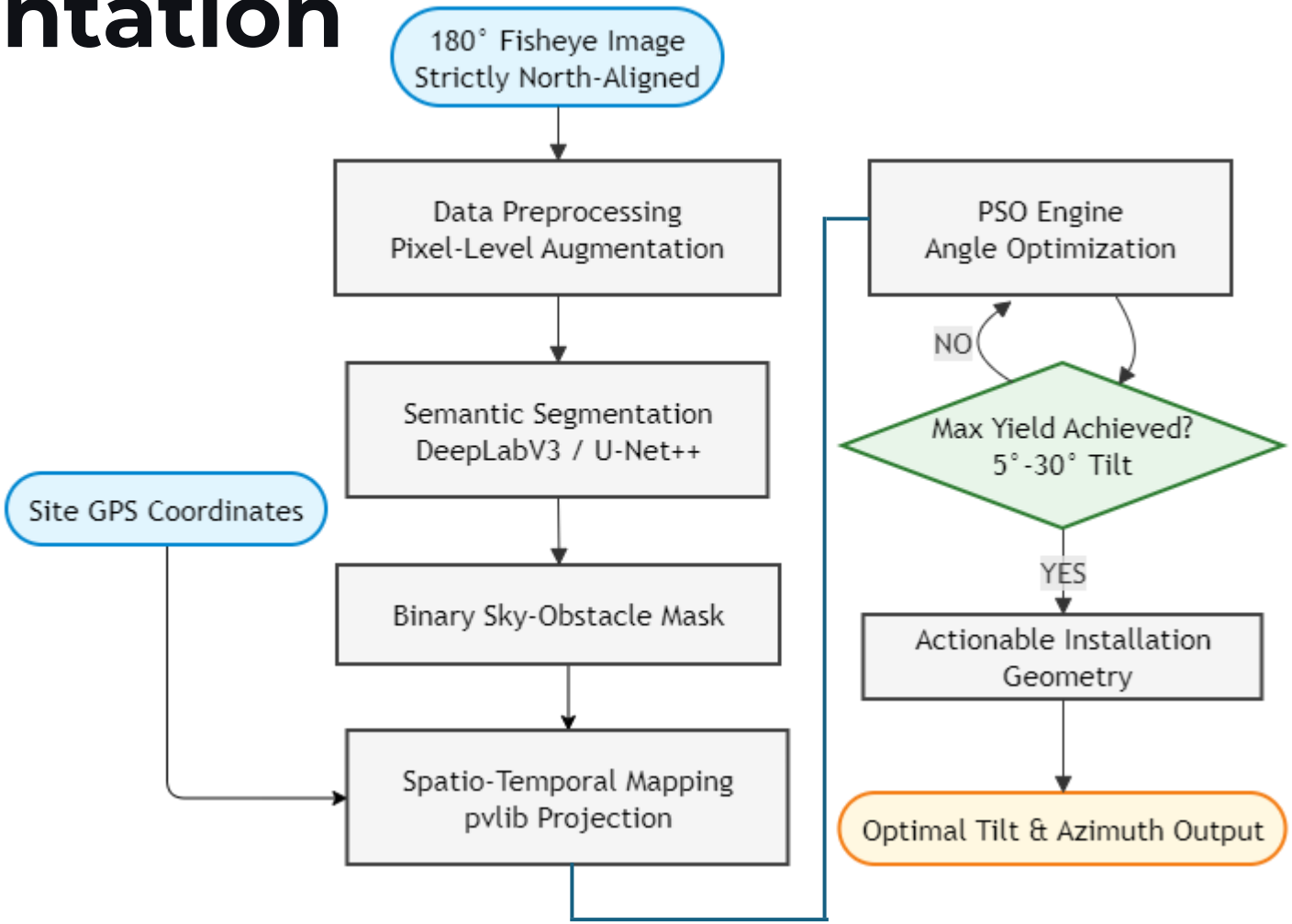
Validation & Evaluation



- **Held-out Segmentation Accuracy:** mIoU and Pixel Accuracy against ground-truth hand-labeled fisheye shading masks.
- **Baseline Ground-Truth Validation:** RMSE & MAE calculated by comparing our pvlib baseline simulation against actual historical inverter yield logs (CSV) from 3-5 existing local installations: By proving that our system's simulated energy yield perfectly matches the actual inverter logs of a standard 7-degree installation, we can confidently guarantee that the increased power output predicted by our AI's newly optimized angles is completely accurate and trustworthy.
- **Comparative Energy Yield Gain:** Demonstrated percentage increase in simulated annual generation (kWh/m^2) by comparing the PSO-optimized geometry against the validated Sri Lankan standard (fixed 7° tilt).
- **Optimization Robustness:** Constraint satisfaction rate, ensuring the algorithm strictly maintains the hardware mounting limits (5°–30° tilt).

Vision-Based Shading Analysis & Solar Geometry Optimization

Implementation Details



Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning for Solar Home Energy Systems



IT23334038 | Wijewardena P.K.L

Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning for Solar Home Energy Systems



Novelty

Problem

- CEB scheduled load-shedding cuts grid power at predictable but battery-blind hours
- Existing home battery control uses static threshold rules (charge to 90%, discharge to 20%)
- No awareness of an upcoming outage battery can be near-empty exactly when it's needed most
- Existing DRL home-energy research optimizes for cost, not blackout survival, and often assumes smart-meter/inverter telemetry unavailable to most households here

Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning for Solar Home Energy Systems



Novelty

Main Objective

- Develop and evaluate a hardware-free deep reinforcement learning agent that makes hourly self-consume / export / reserve decisions for a residential solar-battery system to minimize unserved critical load during scheduled outages, while remaining cost-aware.

Sub Objectives

- Build a custom Gymnasium environment simulating SoC, PV generation, load, and outage state
- Formulate the problem as an MDP with an outage-economics reward function
- Train and compare PPO and DQN agents against a rule-based baseline
- Evaluate on Total Annual Hours of Lost Critical Power + a financial savings metric
- Implement a user-selectable resilience preference (savings / balanced / resilience) via multiple pre-trained policies

Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning for Solar Home Energy Systems

Research Methodology



Simulation-based experimental research, using a custom Gymnasium reinforcement learning environment. This approach is justified because live deployment on real household inverters is infeasible at this scope, and simulation allows controlled, repeatable comparison against a rule-based baseline under identical conditions

Technology used:

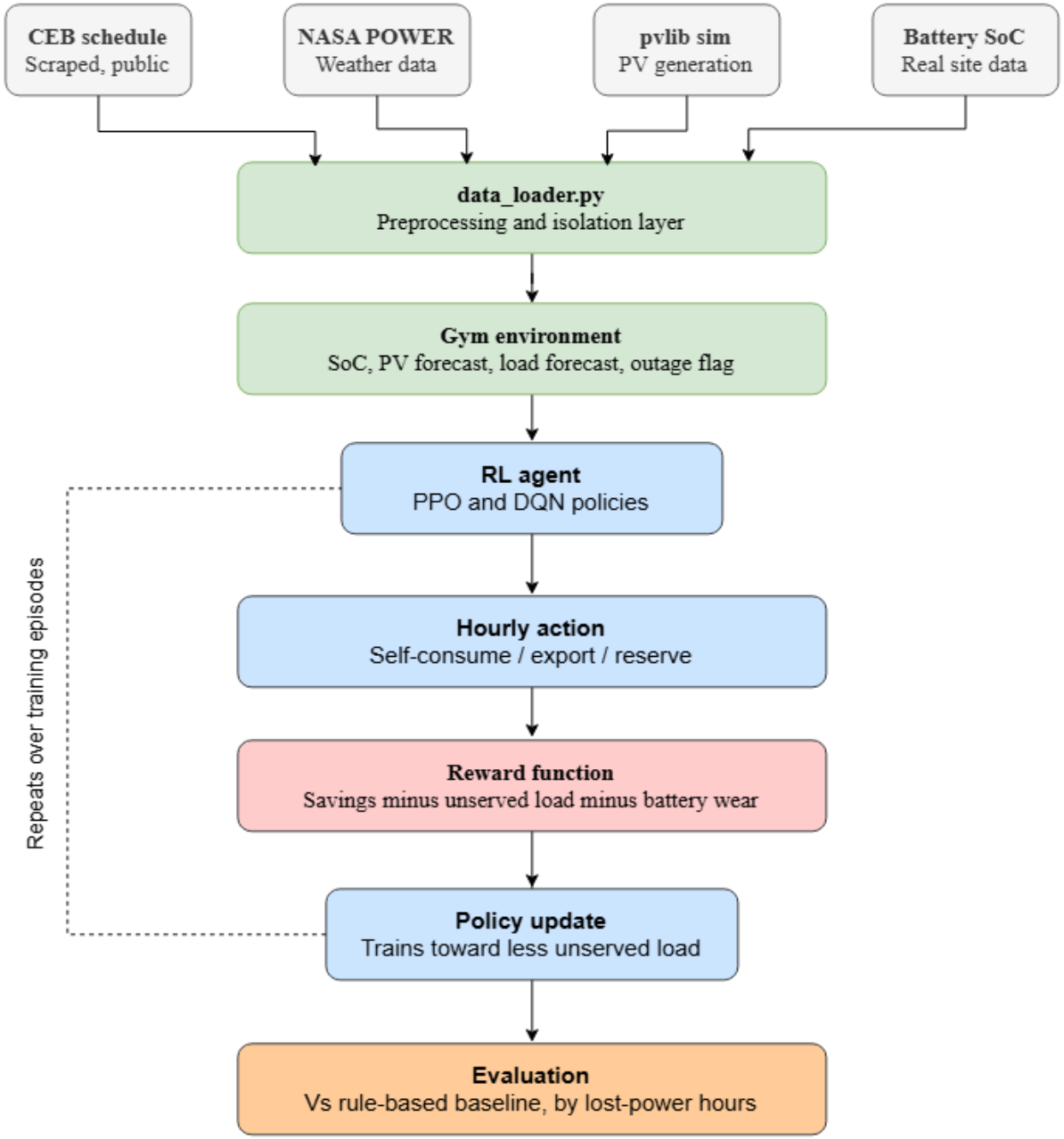
- Reinforcement Learning – PPO(Proximal Policy Optimization) and DQN (Deep Q-Networks)
- Pvlib – physics based PV generation simulation
- NASA POWER API - weather data
- Custom MDP setup - tracks battery level and forecasts, chooses self-consume / export / reserve, and rewards decisions that avoid blackouts while saving cost

Data availability & ethics:

- Real solar company data: PV generation, load curves, battery SoC
- Public data: CEB load-shedding schedules (scraped), NASA POWER weather
- No personally identifying household data used

Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning for Solar Home Energy Systems

Research Methodology



Proactive, Hardware-Free Grid-Survival Optimization via Deep Reinforcement Learning for Solar Home Energy Systems

Implementation Details



System Overview:

The component consumes standardized inputs (schedule, weather, generation, SoC) through a `data_loader.py` interface, feeds them into the Gym environment, and outputs an hourly action recommendation from the trained policy fully hardware-free, requiring no smart meter or inverter integration.

Dependencies:

- Fully independent - own data collection (real company data + public sources) and own PV simulation (pvlib)
- Output format designed to optionally feed a shared household dashboard in future, but functions standalone without one

Commercialization Investment /cost and how to recover this

Ability of Commercialization

A software-based residential solar intelligence platform for homeowners, solar installers and solar service providers.

Investment/Cost

Cost Area	Requirement
Software Development	Web/mobile platform, backend, AI integration
Cloud Infrastructure	Hosting, database, model/API services
Data	Weather/API services where required
Image Acquisition	Smartphone-compatible fisheye lens
Maintenance	Model updates, hosting, support
Integration	Solar installer/inverter integration where applicable

Commercialization Investment /cost and how to recover this

Ability of Commercialization

A software-based residential solar intelligence platform for homeowners, solar installers and solar service providers.

Recovery

Business to customer B2C: For homeowners

- Freemium / subscription

Business to Business B2B: Solar Installation Companies

- Annual software license / pre-installation fee

Enterprise/ Api integration

Shading-Aware AI System for Solar Investment, Health Monitoring, and Grid-Resilience in Sri Lanka

Target market, customer profile, value proposition, intellectual property

Target Market

- **Primary:** Residential solar installation & service companies
- **Secondary:** Residential rooftop solar owners
- **Future:** energy management/ smart home service providers

Customer Profile

- Solar companies
- Homeowners

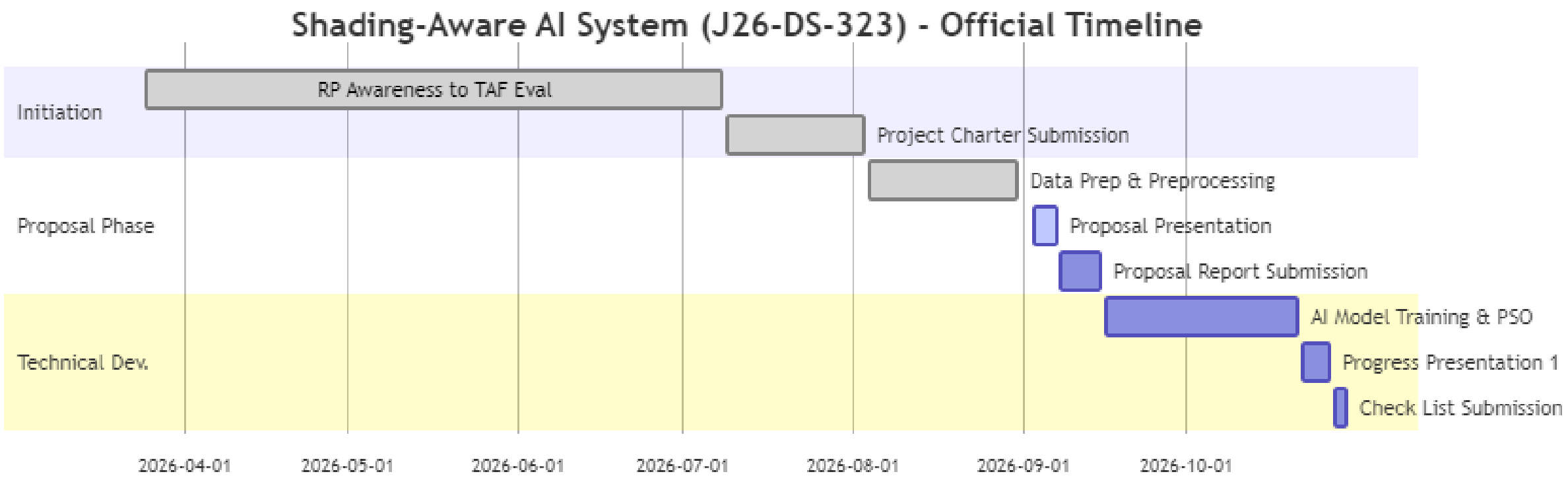
Value Proposition

Optimize Installation
Vision-based shading and panel-angle recommendations.
Monitor Panel Health
Detect tropical degradation and provide interpretable health assessment.
Improve Solar Utilization
Schedule flexible household appliances around available solar generation.
Improve Energy Resilience
Optimize solar/battery usage around expected outages.

Intellectual Property Rights (IPR)

- Proprietary software platform and system architecture
- Trained AI/ML model implementations
- Custom optimization/rescheduling algorithms
- Original datasets
- Software source code protected by copyright

Shading-Aware AI System for Solar Investment, Health Monitoring, and Grid-Resilience in Sri Lanka



Work breakdown structure

1. Installation Optimization	2. Panel Health Assessment	3. Dynamic Appliance Scheduling	4. Grid-Survival Optimization
Vision-Based Shading & Panel Placement	Vision-Based Degradation Analysis	Solar-Aware Energy Utilization	Outage-Aware Energy Management
1.1 Fisheye rooftop image collection	2.1 Tropical degradation image collection	3.1 Day-ahead PV forecasting	4.1 Load-shedding & weather data collection
1.2 Obstacle/sky segmentation	2.2 Defect/degradation detection	3.2 MILP appliance scheduling	4.2 PV/battery/load simulation
1.3 Sun-path modelling	2.3 Panel health scoring	3.3 Forecast-deviation monitoring	4.3 DRL energy-management agent
1.4 Tilt & azimuth optimization	2.4 Maintenance recommendations	3.4 Event-triggered rescheduling	4.4 Resilience strategy evaluation

END.